Multi-digit addition: one model from many

Recognition, associations, problem solving, and memory, composed and proved

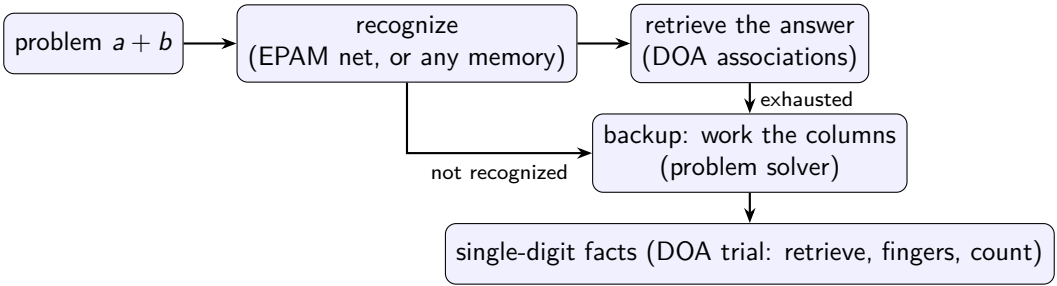
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The task, and what a model must explain

- Children and adults solve problems like $150 + 100$ and $478 + 256$.
- Some problems are *familiar*: the answer is retrieved. Others are worked out, column by column.
- Errors come in kinds:
  - **confident errors**: a familiar-looking problem answered with a neighbour's sum, with no hesitation;
  - **column errors**: a single-digit fact wrong in one column;
  - **procedural bugs**: systematic errors from a procedure with a missing step (no carry).

The architecture



Every arrow is a composition of probability distributions; every box is a theory formalized on its own, with its own theorems.

Component 1: recognition (EPAM)

- A **discrimination net** tests features of the problem's code, and sorts it to a stored image.
- Learning adds a test only where two codes are confused.
- A net that has never needed to tell two problems apart can't: $150 + 100$ is sorted to $100 + 100$ when the net tests only the digit where the familiar problems differ.

Proved for every net: recognition depends only on the features tested; a confusion agrees on every tested feature; learning a problem never disturbs another.

Component 2: distributions of associations (Siegler & Shrager 1984)

- Each problem has associations of varying strength to possible answers.
- Retrieval samples an answer in proportion to strength; it is stated if above a **confidence criterion**; up to a search length.
- If nothing is stated: backup strategies (fingers, counting fingers).
- Used twice here: for **whole problems** (once recognized) and for **single-digit facts** in each column.

Proved: retrieval is used with probability $1 - (1 - R)^n$; the error share on retrieval trials is the wrong answers' share above the criterion; strengthening the correct answer never raises errors.

Component 3: the column backup as problem solving

- A problem space: add a column (its fact from a DOA trial), write, write and carry, finish.
- **The bridge**: the solver answers exactly as the old fixed column procedure did, for every problem and every fact process.
- **Means-ends analysis** (GPS) works units first, though its goal names no order.
- **Repair theory** (Brown & VanLehn 1980): delete the carry rule, and a column sum of 10 or more is an impasse; its repairs are bugs (no carry: $15 + 17 = 22$).

Component 4: memory as a parameter

- Recognition is any **retriever**: study, then retrieve a distribution over stored items.
- EPAM's net is one; ACT-R's declarative memory (activation, partial matching) is another.
- Three **laws** of memory, determinacy, certainty and stability, are enough for confident errors; EPAM's net has all three, ACT-R has none.

So which errors a learner makes depends on which memory they have, and the laws say why.

How they compose

A trial on problem x:

$$\text{trial}(x) = \text{recognize}(x) \gg\!= (\text{retrieve the recognized problem's answer else backup}(x))$$

$$\text{backup}(x) = \text{add units} \gg\!= \text{write and carry} \gg\!= \text{add tens} \gg\!= \ldots$$

$$\text{add column} = \text{DOA trial on its digits} \gg\!= \text{record the sum}$$

- Each $\gg\!=$ is a two-stage draw: probabilities multiply along a path and add over paths.
- Composition is exact: no simulation anywhere; every number is a theorem.
- Because draws regroup (associativity), a theorem about a component holds inside the whole.

What the whole predicts

Prediction	From
Confident errors: every retrieval of a confused problem is wrong, with no extra backup	recognition (EPAM laws) + sharp associations (DOA)
A column error is a single-digit error moved to its place	problem solver + DOA facts
Carrying adds no error of its own	problem solver (bridge)
Procedural bugs only on carry problems	problem solver without the carry rule
Learning removes a confusion without disturbing other problems	EPAM learning

A worked case

	EPAM	ACT-R
$150 + 100$ answered 200 (the neighbour's sum)	45/46	225/529
$150 + 100$ answered correctly	1/46	383/1058
$15 + 17$, no carry rule, strict critics	22, for every such learner	

- Values are exact, at settings chosen for illustration; the predictions above hold for every setting.
- With EPAM, the confusion is certain; with ACT-R, it is graded.

Is the formalization faithful?

- A prose specification, with no Lean in it, and preregistered predictions.
- A fresh AI agent, given only the prose, wrote a Python simulator.
- **No failures**: 243 outcome frequencies on 4 instances matched Lean's exact values; every applicable prediction passed.
- It also found five vacuous predictions and a gap in the formalization, since fixed or recorded.

The blind check predates the problem solver; by the bridge, its results carry over unchanged.

What each component contributes

Component	Source	Role in the whole
Discrimination nets	EPAM (Feigenbaum; Simon)	which problems look alike; certain confusions
Distributions of associations	Siegler & Shrager 1984	answers, confidence, backup choice; single-digit facts
Problem solving	GPS; Brown & VanLehn 1980	the column procedure; its order; its bugs
Retrievers and laws	ACT-R (Anderson); laws new here (the theory matrix)	which memory gives which errors, and why

Each was formalized and proved on its own; the composition is proved too.

Limits, and next

- Worked cases use chosen settings; priors should come from experience, as Siegler & Shrager's did from counting (open issue).
- "Multi-digit" is three-digit.
- **Chunking** would close the loop: a fact found by counting at a retrieval impasse becomes an association, DOA's learning rule as problem-solving learning.
- Subtraction, borrowing, and repair theory's own bug data.